

Kanak Raj

LinkedIn: rajkanak

Github: github.com/kanak8278

Email: kanak8278@gmail.com

Mobile: +918529145565

Google Scholar

EDUCATION

- Birla Insitute of Technology Mesra** Ranchi, India
 - Integrated Master's of Science - Mathematics & Computing; GPA: 8.88 2019 - 2024

PUBLICATIONS

- Cook-Gen: Robust Generative Modeling of Cooking Actions from Recipes** at IEEE SMC 2023.
- Spatial Field Fusion Network (SFFNet) for Panoramic Dental X-ray Segmentation** at IEEE APSCON 2023.
- K-PERM: Personalized Response Generation Using Dynamic Knowledge Retrieval and Persona-Adaptive Queries** at AAAI Spring Symposium 2024.

EXPERIENCE

- Thomson Reuters Lab** Bangalore, India
 - Applied Research Scientist Aug 2024 - Current
 - Legal AI Reasoning and Model Enhancement:**
 - * Pioneered a legal synthetic data generation pipeline for Process Reward Models (PRMs), creating domain-specific training datasets that improved legal reasoning capabilities on LegalBench benchmark tasks.
 - * Evaluated different Test-Time Scaling paradigms for legal reasoning and optimizing inference-time computation.
 - * Architected an IRAC (Issue-Rule-Application-Conclusion) Knowledge Graph framework using Thomson Reuters' Westlaw corpus and court case data, generating high-quality preference datasets that improved legal reasoning alignment in fine-tuned LLMs.
 - AI-Powered Legal Document Update System:**
 - * Build a comprehensive end-to-end LLM Workflow pipeline to update Word Document with XML parsing and intelligent contextual mapping for paragraph identification and edit, achieving 70%+ success rate while preserving complete document formatting integrity.
 - * Engineered human-in-the-loop validation interface with reasoning chains and alert-point mapping, resulting in 60% reduction in manual processing time while maintaining legal accuracy through strategic human oversight and audit trails.
 - DocEvolver:** Created an MVP for "Cursor for Word" as an Extension for updating and understanding MS Word Files for Lawyer-Editors
 - Search-and-Replace Agentic System:**
 - * Architected a multi-agent AI system for automated Word editing with comprehensive validation pipeline (schema enforcement, content integrity, audit logging), achieving 98% accuracy, resulting 65% reduction in manual content revision.
 - * Constructed error-resolving agents with function calling and multi-turn reasoning to fix XML issues, implementing few-shot learning and self-healing mechanisms, resulting in widespread adoption across teams, across hundreds of documents processed monthly with sub-8 second processing time per section.
 - Additional Tools:**
 - * **Truth Social Monitor:** Created a monitoring system for Trump's Truth Social posts with sub-3 second latency, generating automated alerts for Reuters journalists
 - * **Page Flipper:** Revived version of the Page Monitor extension for website tracking that eliminating Visual Ping subscriptions for the team.
 - * These tools provided Reuters with a critical competitive advantage over competitors.
- Microsoft Research** Bangalore, India
 - Research Intern (LLMs & AI4Code) Jan 2024 - July 2024
 - Programming with Representations (PwR):**
 - * Led backend development for Microsoft's PwR Studio platform, focusing on the Natural Language to Domain Specific Language (NL2DSL) translation system using GPT 3.5 & 4, and developed a symbolic translation pipeline that generates finite state machines structured as custom DSL, achieving 85% reduction in hallucinations.
 - * Formulated Rubrics and Evaluations with error correction over DSL which resulted in valid DSL generation increasing from 65% to 95%
 - Jugalbandi-Studio-Engine:**

- * Architected Python-based platform that converts DSL into scalable finite-state-machine-based chatbot applications. This framework reduced the development time by 80% and increased accessibility for government agencies and NGOs.
- * The platform was featured in Satya Nadella's keynote talks, and I represented Microsoft Research in the pilot project, enabling 15+ nontech organizations to develop AI-powered conversational bots.
- o **Jugalbandi(JB) Manager:** Established a chatbot management platform supporting multiple channels (WhatsApp, Telegram, Web) with multilingual text and voice capabilities; integrated Bhashini Speech models with Azure service failover mechanisms, ensuring 70% faster deployment of new chatbots
- o **Open Source Contributions:** We open-sourced our complete system which has been picked by Bhashini to build Chatbots: PwR-Studio, PwR-NL2DSL, Jugalbandi Studio Engine, Jugalbandi Manager.

- **AI Institute, University of South Carolina** Remote
Research Intern (Neuro-Symbolic AI & NLP) Dec. 2022 - April 2024
 - o Master thesis on **Knowledge Enabled Multimodal Ingredient Substitution**, built **knowledge graph** incorporating 27K ingredients and 80K substitution pairs. Enabled precise ingredient recommendations using multimodal and constraint-based searches. We built an LLM-based query module for the ingredient substitution knowledge graph. We also submitted our work to AAAI-25.
 - o Formulated cross-modal Recipe Retrieval and implemented cooking action recognition for recipe analysis, achieving **95% Recall** score leading to paper **Cook-Gen** at IEEE SMC'23 under Dr. Amit Seth.
- **Saarland University** Germany
Visiting Researcher May 2023 - Aug 2023
 - o Conducted research in **Satellite Image Superresolution**, enhancing **disaster analysis** through high-temporal frequency & low-resolution images using GAN & Diffusion models under Dr. Ingmar Weber and Dr. Ferda Ofli.
- **University of Maryland** Baltimore
Research Intern (Natural Language Processing) Oct. 2022 - April 2023
 - o Worked on developing **Personalized Response Generation** models using **reward scaling** over BART & T5, accepted at AAAI Symposium'24 and improved NUBIA score by 10% under Manas Gaur.
- **EdgeNeural.ai** Remote
AI Engineer Intern June 2022 - Aug 2022
 - o Engineered **Training** and **Inference** pipeline, Optimized and benchmarked DL models(YOLO, ResNet, BERT, UNet) with **Quantization** and **TensorRT**, achieving **10-15x** speedup in latency.
- **AI MAGE (WETHEKOO)** Remote
Computer Vision Engineer March 2022 - April 2022
 - o Orchestrated the creation of a **Fashion Tagging Engine** using fine-grained **Apparel Segmentation** (iMaterialist), performing **benchmarking** like FPS, inference time and memory usage.
- **Rhizicube Technologies** Remote
Software Engineer Intern June 2021 - Sept. 2021
 - o Constructed **Server & REST APIs** in Golang(Gin) and built a **Kafka** based real-time data streaming pipeline for a Consumer Data Platform. Independently programmed LinkedIn and Organization scrapers using Selenium for financial data and reports.

SKILLS

- | | |
|--------------------------------|---|
| • General | Python, C/C++, SQL, Docker, Kafka, Git/GitHub, AWS, Selenium, BeautifulSoup |
| • Web Technologies | HTML/CSS, Flask, JavaScript, Beego/Gin(Golang) |
| • Data Science & ML | Tensorflow/Keras, PyTorch, Scikit-Learn, OpenCV, OpenVino, TensorRT, SpaCy |
| • Generative AI | Large Language Models, OpenAI, Anthropic, Gemini, RAG, LlamaIndex, LangChain, Agentic Systems |

LEADERSHIP & VOLUNTEER EXPERIENCE

- Served as **Joint President** and formerly **Joint General Secretary** of **Society for Data Science**, BIT Mesra. Led a community of 150+ members in eastern India's largest Data Science Summits in 2021 and 2022, attracting 450+ attendees and 15+ industry speakers with 40% year-over-year growth.
- **Instructor** in several workshops, including Python and Introduction to Data Science, Machine Learning, Introduction to Deep Learning, Git/GitHub, and OpenSource. Conducted 10+ technical sessions for 250+ students with 92% satisfaction rate and mentored 20+ students to develop practical skills.